

DANIEL: A Deep Architecture for Automatic Analysis and Retrieval of Building Floor Plans

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Abstract—Automatically finding out existing building layouts from a repository is always helpful for an architect to ensure reuse of design and timely completion of projects. In this paper, we propose Deep Architecture for fiNdIng alikE Layouts (DANIEL). Using DANIEL, an architect can search from the existing projects repository of layouts (floor plan), and give accurate recommendation to the buyers. DANIEL is also capable of recommending the property buyers, having a floor plan image, the corresponding rank ordered list of alike layouts. DANIEL is based on the deep learning paradigm to extract both low and high level semantic features from a layout image. The key contributions in the proposed approach are: (i) novel deep learning framework to retrieve similar floor plan layouts from repository; (ii) analysing the effect of individual deep convolutional neural network layers for floor plan retrieval task; and (iii) creation of a new complex dataset *ROBIN* (Repository Of BuildIng plaNs), having three broad dataset categories with 510 real world floor plans. We have evaluated DANIEL by performing extensive experiments on *ROBIN* and compared our results with eight different state-of-the-art methods to demonstrate DANIEL's effectiveness on challenging scenarios.

Index Terms—Floor Plan; Retrieval; Deep Learning; CNN

I. INTRODUCTION

Exponential growth of population and rapid urbanization has caused huge boom in the real-estate industry. According to the annual report in [1], nearly 92% of home buyers are using online platform to search for homes, and 82% of them prefer search by example image. To balance this demand and supply chain, while designing a new floor plan, architects often refer to existing projects. An architect comprehends the buyers requirements, recalls a similar floor plan, and manually finds it from the previously designed projects. Looking into each and every floor plan manually is a tedious task and an artificially intelligent system for automatic understanding and retrieval of floor plans is essential to solve this problem.

Apparently, a canonical Content Based Image Retrieval (CBIR) based solution seems unrealistic in this case as the same will not capture the interior details of a house. The buyers may want to have a house with “1 master bed room, 2 smaller bedrooms, kitchen on the southern part of the house, with a garage to right side of the entry...”. In such a scenario, mere external similarity is not sufficient, and interior designs need to be looked into. This situation steered us to choose a floor plan

image, instead of an image of an entire house or rooms for query. A floor plan image captures the horizontal cross section of a house, as well as its interiors. Analysing the floor plan image and finding out similar floor plans is thus, an important and challenging task, however, less explored in the past.

In literature, work related to retrieval of similar floor plans from a repository is rare. In [2], [3], a preliminary model for floor plan retrieval task was introduced that works on a small dataset. Existing works in the literature contain several other applications of floor plan analysis. Examples of other related work include (i) generating 3D models [4]; (ii) detecting rooms and their connectivity topology [5]; and (iii) sketch based retrieval [6]. In this paper, we propose DANIEL (Deep Architecture for fiNdIng alikE Layouts), a deep neural network based, artificially intelligent (AI) search framework. DANIEL automatically analyses the floor plans accurately and is capable of finding out alike floor plans from a large repository.

To the best of our knowledge, *DANIEL is the first AI system that applies deep learning approaches to automatically analyze building floor plan images and retrieve similar plans from a large-scale repository.* Unique contributions made in this paper are: (i) Proposal of a distinct, versatile dataset *ROBIN* (Repository Of BuildIng plaNs), dedicated to the task of floor plan retrieval; (ii) A deep learning framework for extraction of both low and high level features for the task of fine-grained retrieval; and (iii) Analysis of the effect of individual deep convolution neural network hidden layers on retrieval results. Our proposed technique can find application in online property sale/rent scenarios where the buyer has preferred features related to the room semantics.

Organization of the paper: A brief description of the related works is given in Sec. II. Overview of DANIEL and *ROBIN* dataset is discussed in Sec. III. Section IV describes the proposed architecture of DANIEL. Experimental results and system evaluations are given in Sec. V. Finally, Sec. VI summarizes our findings and concludes this study.

II. RELATED WORK

A very closely related work to ours is the method proposed in [2], which is on semantic retrieval of building floor plans. The method proposed a spectral embedding method to match a pair of floor plans. In [7], challenges of using graph matching technique for architectural floor plan retrieval was shown. A

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spectral method was proposed in [8]. The method decomposed graphs into non-overlapping neighbourhoods and used them for the purposes of both matching and clustering. Another related method is the one proposed in [6], where a sketch-based similar floor plan retrieval approach was proposed to query the floor plan repository by categorizing areas in a floor plan as rooms, zones, units and levels.

Earlier structural analysis in architectural layouts were explored. The work on floor plan analysis [9] used a simple technique for creating CAD models from hand sketched floor plans. This work was followed by method proposed in [10], which deals at applying techniques like Hough transform and subgraph isomorphism for floor plan interpretation. An end-to-end system is proposed in [11] that extracts the room labels to identify the functions of the rooms for the analysis of architectural diagrams. Differentiation between thick, medium, and thin lines and the removal of components outside the convex hull of the outer walls for segmenting the room layouts in an efficient manner was proposed by [12].

Several generic image retrieval systems were also proposed in the past [13], [14], where features like Histogram of Oriented Gradients (HOG) [15], Bag of Features (BOF) [16], Local Binary Pattern (LBP) [17] and Run-Length Histogram [18] has been used. Online Algorithm for Scalable Image Similarity (OASIS) [14], proposed a method for learning pairwise similarity between pair of images for image retrieval task. In [13], an object-centric CBIR has been proposed which tries to simultaneously optimize “*What and Where*” problems, for image retrieval tasks. All these techniques have shown results on object-centric databases.

In contrast to retrieval, symbol spotting is another important area explored earlier. Pattern sensitive, moment invariant features like Zernike moments have been proposed in [19] for symbol spotting. In [20], a graph path hashing of the shape descriptors has been proposed to minimize the search space using Hamiltonian paths while spotting the symbols. Alternatively, in [21] an approximate Edit Distance algorithm and near convex grouping approach was explored to create region adjacency graph on graphical documents for symbol spotting. Rotation and translation invariant features for symbol spotting in documents were proposed in [22], and [23]. However, most of these techniques were applied on small datasets, and the models discussed above were “shallow models”, which are unsuitable for handling large, complex datasets.

III. SYSTEM OVERVIEW AND DATA SOURCE

In this section, we give a brief overview of the proposed DANIEL framework for similar floor plan retrieval. We also discuss the rationale behind the proposal of the ROBIN dataset and give the details of its key features.

A. Brief description of DANIEL

Figure 1 depicts the complete framework of our proposed technique. The entire framework has been divided into two main phases. They are: (i) deep feature representation; and (ii) matching and retrieval. DANIEL first learns the deep

feature representation for floor plan retrieval task using a few samples from the proposed ROBIN dataset. The learned deep representation helps to extract deep features from the samples of ROBIN, and stores it in the feature database. During the retrieval stage, DANIEL extracts the deep features from query image also using the same deep representation. The extracted deep features are then matched with the deep database features. A similarity score for a particular layout is thus calculated for all the query samples. To understand the effectiveness of the individual layers or feature representation of deep network, we have also calculated the similarity using all the individual deep features/layers. Details of the layerwise analysis of DANIEL are discussed in Sec. V-C.

B. Features of the ROBIN Dataset

Symbol spotting and room segmentation in floor plan images has been explored in the past. For that purpose two public dataset were proposed. They are: (i) SESYD [24] and (ii) CVC-FP [25]. The former has 10 classes of floor plans, where there are 100 samples/class. The latter has 122 scanned floor plan documents divided into 4 categories based on the origin and style. In [2], authors have introduced the problem of floor plan retrieval. However, they performed experiments on SESYD dataset, which is inappropriate for application to a retrieval task. Also, in the CVC-FP dataset, the samples are insufficient in number for the task of floor plan retrieval.

Keeping in mind the current status of floor plan analysis research, we believe that a publicly available benchmark dataset will aid research for the community in this particular area. Therefore, in this paper we propose a new floor plan dataset, *ROBIN*, which consists of 3 broad categories of layouts. Figure 2 depicts a snapshot of the various categories of dataset in ROBIN. The unique characteristics of ROBIN is that the dataset is designed keeping in mind the need of a potential buyer. Every prospective buyer wants to have certain amenities and functionalities in their house. Thus, in the dataset, there are three broad categories, which are different from each other in terms of the number and type of rooms present in a floor plan. Each broad category is further classified into 10 sub-categories depending upon the global layout shape of the floor plan. We have followed standard notations given in [24], for all the floor plans so that ROBIN can be used for other floor plan analysis tasks as well, and not only for retrieval. ROBIN helps in better visualization of the floor plans and aids in efficient capturing of various high-level features while fine-grained retrieval. To facilitate research in similar floor plan retrieval, we have made the ROBIN dataset publicly available [26].

IV. PROPOSED DEEP LEARNING ARCHITECTURE

Convolutional Neural Networks (CNNs) have shown promising results for various computer vision tasks like classification, and detection. However, researchers as yet, have not explored, the performance of CNNs for floor plan retrieval tasks. In this paper, our goal is to come up with new effective deep learning based feature representation and to investigate the effectiveness of learned features on floor plan retrieval

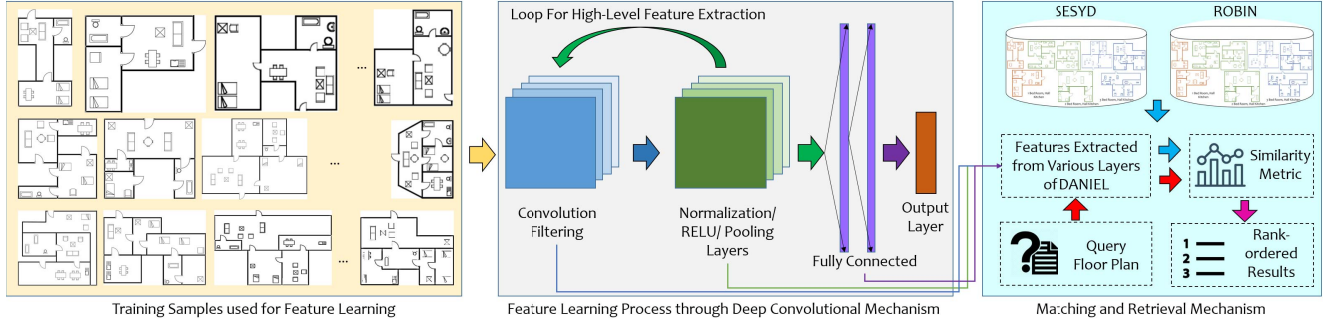


Fig. 1. Architecture diagram of DANIEL: a Framework of Deep Learning for automatic analysis and retrieval of floor plan images.

tasks. In particular, we want to address two open issues: (i) *how to learn new efficient deep learning feature representation for floor plan retrieval task?*; and (ii) *how the individual deep feature layers will affect the performance of the retrieval system?* Our proposed architecture has two main components:

A. Feature Representation

Traditionally, feature extractors are manually engineered and optimised through a laborious trial-and-error cycle involving adjusting the features and re-learning the classifiers. In this work, we propose to learn the representation using a CNN instead, by jointly optimising the performance of the features, as well as, of the classifiers. Existing literature also validates that deep features are a better way to represent an image. However, it is also shown that improper training of deep layers may lead to over-fitting and under-fitting of a model which can serve as a disadvantage.

In this work, we have used a combination of convolutional, pooling, normalization, RELU, drop out and fully connected layers (see Fig. 3) similar to described in the work AlexNet [27], to obtain an effective feature representation from floor plan images. Our model is different from AlexNet in terms of two properties: (a) not training with the relighting data-augmentation; and (b) the order of pooling and normalization

layers is switched (pooling is done before normalization) [28]. For learning a representation of CNN (deep) layers, we have used a few samples from proposed floor plan dataset. Unlike others, we took the activations of all the layers (convolution, pooling, norm, and fully connected) to investigate the effectiveness of each individual layer (see Fig. 1). The major motivation behind using lower convolution neural network layers is to understand the effectiveness of low level features as compared to higher-level feature representation for floor plan retrieval tasks.

To obtain the feature representation from query as well as the database samples, we directly feed the images into the input layer of the learned CNN model, and then take the activation values from all the layers¹ (see Fig. 1). Since we only need to compute the feed-forward network based on the matrix multiplication for one time, the whole scheme is highly time efficient. In matching and retrieval stage, query samples are matched with database samples to effectively retrieve the rank-ordered samples from the database.

B. Matching and Retrieval

Floor plan layout matching using the extracted features is essential for retrieving similar layouts to the query layout. In the rest of this section, we describe the procedure to calculate matching score M for a pair (query and database) of layouts, corresponding to learned deep features.

Let N be the total number of images in the form of (x_ρ, y_ρ) , $\rho = 1, 2, \dots, N$ where, x_ρ is the observed variable, y_ρ is the corresponding class label in a retrieval database. Let $F(x_\rho) : \{F_1(x_\rho), F_2(x_\rho), \dots, F_L(x_\rho)\}$ be the set of hidden layer features extracted for x_ρ sample, where $F_p(x_\rho)$ represents p^{th} hidden layer feature extracted from x_ρ sample in database and L represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, we have extracted features from 12 hidden layers of CNN model (see Fig. 3). For each sample in the database, features from each hidden layer ($F_p(\rho)$) are extracted and stored in the feature database as discussed in Sec. IV-A. Let $F(q) : \{F_1(q), F_2(q), \dots, F_L(q)\}$ be the the set of hidden layer features extracted from a query image (q). In this work, we have performed two different types

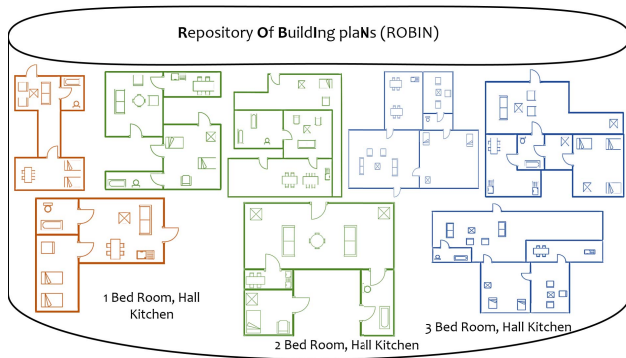


Fig. 2. A visual illustration of proposed ROBIN Dataset. Various categories of rooms are highlighted with different colors.

¹For conv, norm, and pool layers, we have vectorized the activation output

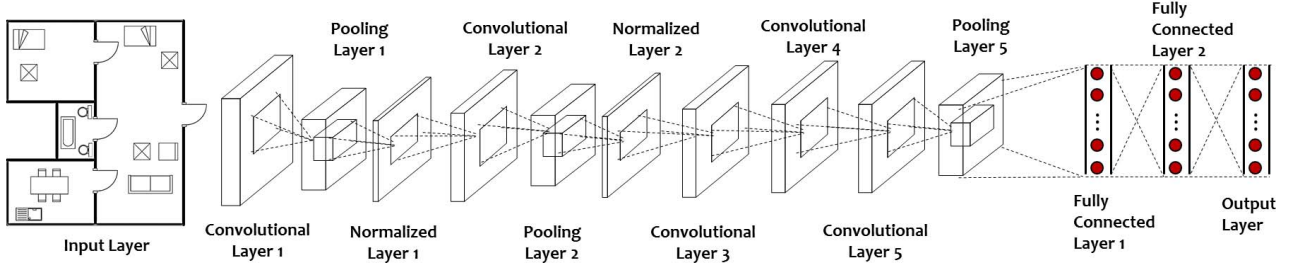


Fig. 3. Convolutional Neural Network (CNN) feature representation (Layers)

of analysis within our proposed framework, using an extracted features from all the hidden layers:

- **Similarity based on each learned hidden layer feature representation:** In this approach, we perform extensive analysis by comparing the performance of individual hidden layer for proposed floor plan retrieval task. The Matching Score (M) between query image (q) and an image sample from the database (x_p) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M = \sum_{i=1}^{|F_p(x_p)|} |F_p^i(x_p) - F_p^i(q)|_2 \quad (1)$$

- **Similarity based on combination of hidden layer feature representation:** We try to find best hidden layer representation for each retrieval sample by combining the matching score outputs of each hidden layers using a $\min$ operation. The Matching Score (M) between a query image (q) and a database image (x_p) using all the hidden layer features is calculated (and used for ranking), as:

$$M = \underset{p}{\operatorname{argmin}} \sum_{i=1}^{|F_p(x_p)|} |F_p^i(x_p) - F_p^i(q)|, \quad p \in (1, \dots, L) \quad (2)$$

V. EXPERIMENTS AND RESULTS

We performed our experiments on SESYD [24] and ROBIN to show the effectiveness of DANIEL. In [24], there are 10 different types of layouts, each with 100 images varying in their arrangement of furniture inside the rooms. The image sizes in the database vary from 6775×2858 to 2056×1837 . All floor plans are binarized ensuring structural information usage while analysis. Details of ROBIN are given in Sec. III-B.

A. Experimental Setup

We have split ROBIN into two sets: (a) training - comprises of 30% samples and used for deep feature representation; and (b) testing - comprises of rest 70% samples and used as test (query) samples. As a retrieval database, we have used entire ROBIN dataset. Each floor plan image has been resized to a size of 512×512 for both matching and feature representation task. For deep model configuration, we have

used the architecture described in Fig. 3, with same parameters as discussed in [27]. The same model pre-trained on ROBIN is used for feature representation of the SESYD dataset.

B. Visualization of Results

DANIEL is able to successfully retrieve a rank order set of floor plan images using the proposed deep learning framework. The top 5 rank ordered retrieval results for three different queries on ROBIN, are shown in Fig. 4. The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with “red” color. For all the results shown in Fig. 4, we have used the features extracted from the second Normalization layer of the CNN stack, as it yields the best performance. The rationale behind selecting the normalized layer feature is discussed in details in the next subsection. The first rank ordered result retrieved from the layout database is the query image itself. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 4 (a), all the top 5 samples belong to the query class. Global layouts of the all the floor plans are identical and so is the number of rooms in the floor plans. The individual plans differ in terms of the number of furniture present in the rooms and their position. Readers are requested to note the difference in the shape and position of the individual rooms also for the rank ordered samples. There are 4 rooms in each floor plans, however the DANIEL is able to rank order the samples correctly by ordering the most similar ones to the query first, followed by the rest.

Figure 4 (b-c) depicts some of the failure cases. In Fig. 4 (b), the 4th ranked result is erroneously retrieved (highlighted by “red”) by DANIEL. However, the 5th sample is correctly retrieved. Justification of the Rank 4 erroneously retrieved result can be understood through qualitative analysis. E.g., room 1 in the query layout is adjacent to room 2, 3 and 4, that is identical to the adjacencies shared by room 1 in Rank 2 retrieved result. Moreover, the size and type of furniture in the room 1 is same in case of the query and the rank 4 sample. Hence, similarity score between the query floor plan and the retrieved Rank 4 layout is high, as compared to that with the rank 5 result. However, the similarity score of the rank 1-3 samples are marginally high as compared to rank 4 samples and hence they are ranked higher. In contrast, Fig.

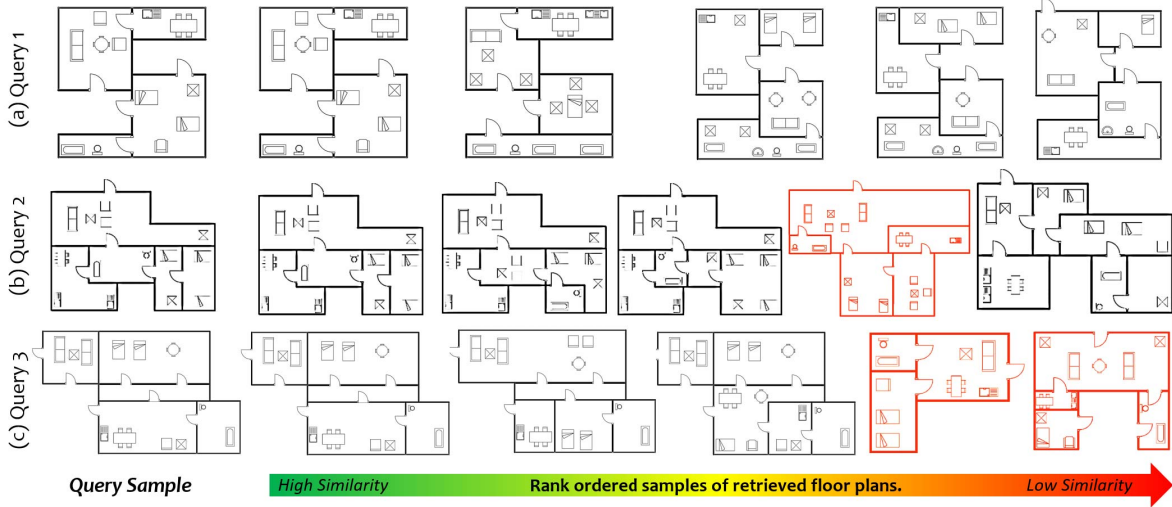


Fig. 4. Rank ordered retrieval result of proposed DANIEL for three different query floor plans from the proposed ROBIN dataset. Here, top five rank ordered floor plans are shown for each query.

4 (c) shows an instance where two out of top five retrieved results are incorrect. The rank 4 image has only three rooms, while the the rank 5 floor plan has four rooms in it. In this case DANIEL is unable to capture the detailed features of the floor plans correctly. Even though, the number of rooms in a floor plan is different than the query, DANIEL ranks it higher.

C. Performance Evaluation

We quantify the performance of DANIEL using a Precision (P) and Recall (R) plot. We have averaged the precision values over all the queries for the particular recall values. Given a query floor plan image, retrieved layouts should belong to the same sub-category of layouts as the query, keeping in mind the preference set by the property seeker during querying. Remember that the global shape of the layout is criteria for class belongingness of a given floor plan. We have shown the P-R plot for only proposed *ROBIN* Dataset, as for *SESYD* dataset our proposed and some of the existing state-of-the-art techniques [2], [3], [15] yielded flat PR curve (Precision value 1 for all Recall values). The reason behind such a flat curve for *SESYD* dataset is the simplicity of the dataset.

To understand the effectiveness of the proposed CNN layers, we compared the PR and Mean Average Precision (MAP) of floor plan retrieval task using each layer separately (see Fig. 5 and Table I). From Fig. 5 and Table I, it can be concluded that: (a) it is not always true that last layer i.e. fully connected layer will always work best; and (ii) normalized and pooling layers are more powerful as compare to convolutional layer for floor plan retrieval task.

To show the effectiveness of our proposed framework, we have also compared our work with several state-of-the-art (SOA) techniques. Figure 6 depicts the PR plot comparing result of DANIEL on ROBIN, with the other SOA techniques. It can be observed that DANIEL with features extracted from

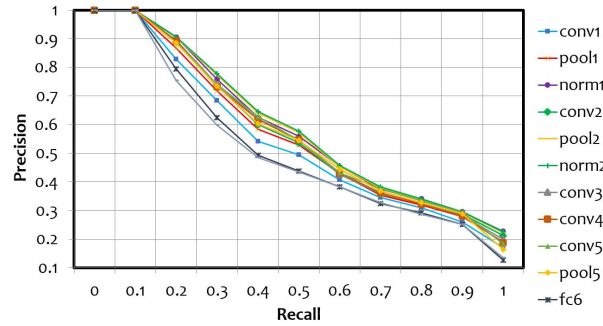


Fig. 5. Precision and Recall (PR) plot, comparing result of DANIEL on ROBIN dataset, using all the hidden layers CNN features.

Layers	conv1	pool1	norm1	conv2
Ours	.503	0.527	0.554	0.537
Layers	pool2	norm2	conv3	conv4
Ours	0.557	0.561	0.539	0.537
Layers	conv5	pool5	fc6	fc7
Ours	0.541	0.536	0.472	0.465

TABLE I
PERFORMANCE COMPARISON (MAP) OF CNN LAYERS USED FOR FLOOR PLAN RETRIEVAL TASK (ON ROBIN DATASET).

the normalized layer 2 (see Fig. 3) performs best. It can also be noted from Fig. 6 that proposed min (which selects best layer for each retrieval sample) based approach is also outperforming all the state-of-the-art techniques. There is a significant difference in the performance while using deep features as compared to other features or techniques.

As illustrated in Fig. 4 (a), given a query sample the rank 1 result is the query floor plan itself. This leads to the highest precision value of 1 during the initial recall. With further

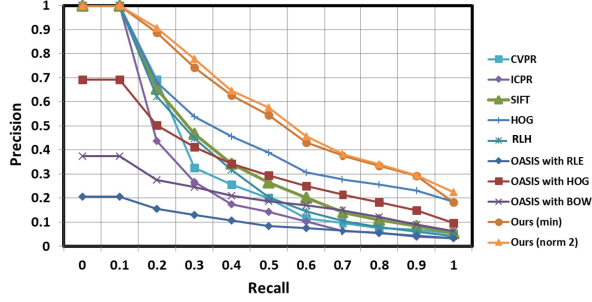


Fig. 6. Precision and Recall (PR) plot, comparing state-of-art methods with proposed DANIEL framework on ROBIN dataset.

Techniques	Results on <i>ROBIN</i>
CVPR [3]	0.29
ICPR [2]	0.23
HOG [15] based CBIR	0.40
SIFT [16] based CBIR	0.33
RLH based CBIR	0.30
OASIS [14] with HOG	0.31
OASIS [14] with BOW	0.19
OASIS [14] with RLH	0.01
Ours (min)	0.54
Ours (norm2)	0.56

TABLE II
PERFORMANCE COMPARISON IN TERMS OF MAP.

retrieval the average precision value decreases due to some incorrectly retrieved results belonging to other categories of layouts as compared to the query layout. It is to be noted that area under the PR curve is the highest for proposed deep learning based retrieval framework, as compared to other techniques applied for floor plan retrieval task.

Reason behind the under performance of OASIS [14] is its focus on the difference of similarity values between relevant and irrelevant image pairs, while ignoring the same between highly similar images. Hence, when the same features (HOG, SIFT, RLH) are used under a canonical CBIR paradigm, they yield superior results than OASIS (see Tab. II). Results justify the importance of the introduction of ROBIN. Although SESYD is used extensively for symbol spotting task, is not suitable for a retrieval task. Characteristics of a challenging dataset like inter-class similarity (overlapping classes) and variation in the samples of the same class is missing in the SESYD. Hence, all the techniques exhibit high accuracy. However, ROBIN was designed keeping in mind the real-world scenario, and hence is more practical.

VI. CONCLUSION

In this work, we have proposed DANIEL, a deep learning framework for floor plan retrieval. DANIEL captures semantic features of floor plans and is able to successfully retrieve them from a large repository. We have also proposed a dataset named ROBIN, to facilitate research on automatic floor plan analysis and retrieval. Extensive experiments were conducted and performance was compared with eight different state-of-

the-art methods to demonstrate DANIEL's effectiveness. In future, we aim to extend the proposed floor plan dataset to better capture deep learned features while training. Also, we aim to apply deep learning techniques to other floor plan analysis tasks such as symbol spotting and perceptual analysis.

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